

SHOU-JEN (LAWRANCE) CHEN

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Summary

- AI engineer with experience building end-to-end intelligent systems, LLM-driven workflows, and practical machine learning applications for real-world use cases. Strong in Python, PyTorch, and rapid prototyping, with a track record of translating technical ideas into usable systems through iteration, evaluation, and strong engineering execution. Motivated by building ambitious AI-powered products that combine technical depth, product intuition, and high craft.

EDUCATION

Electrical Engineering & Computer Science, University of California, Berkeley (UCB) **Berkeley, California**

Master of Engineering, Area of Concentration: Robotics and Embedded Software *Expected Date: Oct 2025 - May 2026*

- GPA: **3.873**/4.0 (Selected Courses: Machine Learning, Embedded Systems, Deep RL, Data Science)

Electrical Engineering, National Taiwan University (NTU) **Taipei, Taiwan**

Bachelor of Science in Engineering *Sep 2020 - Jun 2025*

- Dean's List: 2023 Fall (**Top 5%**) & **NTU Admissions Scholarship** for Overseas Chinese Students

Core Skills: Python, Machine Learning, Data Structures & Algorithms, PyTorch, Reinforcement Learning, C++, MuJoCo, ROS

PROFESSIONAL EXPERIENCE

Software Engineer **Berkeley, California**

JPMorganChase Software Engineering Job Simulation on Forage 🏢 *JPMorganChase* *April 2026*

- Integrated Kafka into a Spring Boot microservice to consume and deserialize high-volume transaction messages using a configurable topic and embedded Kafka test framework.
- Implemented transaction validation and persistence logic with Spring Data JPA and an H2 SQL database, including entity modeling and balance updates across relational User records.
- Connected the service to an external REST Incentive API using RestTemplate and exposed a balance-query endpoint via a Spring controller, returning JSON responses while maintaining clean architectural boundaries.
- Verified system behavior using Maven test suites and debugger-driven inspection, ensuring reliability across message ingestion, database operations, and external API interactions.

Industry/Lab Researcher **Taipei, Taiwan**

Industry-University Collaboration, TrendForce Corporation *Oct 2024 - Jun 2025*

- Developed automated content generation service integrating LLM APIs with structured prompt templates.
- Built LLM-powered text generation workflow with controllable style parameters and automated quality evaluation loops.
- Iteratively refined prompts and model configurations to improve consistency and reliability of generated outputs.
- Implemented style-controlled generation via parameterized prompts and evaluation loops for TrendForce newsletter.

Intern — Software Engineer Team **Taipei, Taiwan**

Software Engineer Internship, Pegatron Corporation *Jul 2023 - Aug 2023*

- Collaborated in a team of three to develop "Visual Prompt Defect Detection" for an internal internship project competition, receiving **1st Place** among four total groups. Our project focuses on using in-context learning.
- Designed UI using Gradio for project demo & built backend data ingestion pipeline storing image datasets to storage via APIs.
- Completed training on **Git, Docker, Kubernetes, Python, Pandas, Numpy, and MongoDB**.

Intern — Software Engineer Webpage Developer **Taipei, Taiwan**

Software Engineer Internship, IADIY Photonics *Jul 2020 - Aug 2020 / Jul 2021 - Aug 2021*

- Designed software enabled auto-driving and basic sensor mapping using Robot Operating System on Linux.
- Assembled robotic devices, including constructing the hardware framework and configuring wires, running on Raspberry Pi.
- Initiated a Kickstarter Campaign in a team of 5 for the product, Genix Lite (air purifier) and designed company website using HTML and CSS.

PROJECTS

Risk-Triggered Latent-Space Corrections for Near-Failure Recovery

Berkeley, California

Reinforcement Learning Researcher, **UC Berkeley**

Feb 2026 – May 2026

- Designed a concurrent decoupled correction architecture for Soft Actor-Critic (SAC) where a lightweight MLP applies additive latent corrections Δz to the actor's hidden representation; a geometric near-failure trigger (sliding-window range check on hand-to-object and object-to-goal distances) gates corrections to only **2–12%** of steps at convergence.
- Identified and resolved co-adaptation via gradient isolation: applied stop-gradient on actor encoder during correction update, preventing encoder from learning latents and ensuring base actor remains independently capable at deployment.
- Evaluated across 4 Meta-World manipulation tasks; improved mean final success rate by **+57pp** on push-v3 (32.5%→89.5%) and **+27.5pp** on pick-place-v3 (66.0%→93.5%), achieving full seed reliability (4/4) versus 0/4 under concurrent training.

AI-driven Personalized Assisted Device

Berkeley, California

Machine Learning Engineer / AI Engineer (Capstone Project), **UC Berkeley**

Aug 2025 – May 2026

- Built a 3-phase LLM-guided optimization pipeline over a 15-dimensional design space, converging to a personalized assistive device design in as few as 2 iterations and achieving **83% overall convergence rate** across 12 runs and 4 participants.
- Designed task interfaces across 8 simulated grasping environments, enabling evaluation of grip force and tasks success rates.
- Optimized designs reduced required grip force to a mean of 2.5 Nm, with **7 of 8 tasks achievable at or below 1.7 Nm**.

IMU-Controlled Quadruped Robot

Berkeley, California

Machine Learning Engineer / Robotics Engineer, **UC Berkeley**

Aug 2025 – Dec 2025

- Implemented a modular 4-stage locomotion pipeline—sinusoidal foot trajectory generation, closed-form planar IK (two-link law-of-cosines solver with workspace clamping), phase-offset gait coordination, and synchronized motor dispatch—supporting **5 gaits** with stride up to 50 mm and maximum forward velocity of **17 cm/s** (>1 body length/s).
- Designed and calibrated trot (50% diagonal phase offset) and walk (25% per-leg offset) gaits; implemented asymmetric stride-length scaling for skid-steer differential turning and a single-cycle jump maneuver with automatic return to neutral stance.
- Debugged and established Raspberry Pi 4 to Dynamixel RX-24F communication over U2D2 UART; synchronized all 8 joints at **100 Hz** with **0.29°** per-tick resolution over a **1 Mbps** daisy-chained Dynamixel bus.
- Built a finite-state-machine command dispatcher routing IMU/EMG gesture outputs to joint trajectories; evaluated SVM (RMS, MAV, waveform-length, zero-crossing features) and DNN classifiers on **8-channel sEMG** at 250 Hz, reaching up to **40%** gesture accuracy and deploying IMU-threshold control for the final demo.

LangGraph ML Research Buddy

Personal Project

Machine Learning Engineer, **Personal Project**

2026

- Built a stateful six-agent LangGraph pipeline (planner, log analyzer, claim checker, reward-hacking auditor, critic, writer) with conditional edges, bounded retry/reflection loops, and a human-in-the-loop checkpoint for ML experiment review.
- Implemented SqliteSaver-backed durable execution enabling mid-run resume, Pydantic structured outputs per node, a custom BaseCallbackHandler tracing every LLM call to per-run artifacts, and a FastAPI/SSE streaming service; provider-agnostic across Anthropic, OpenAI, and Ollama.

RAG-Forge: Modular RAG Pipeline Framework

Personal Project

Machine Learning Engineer, **Personal Project**

2026

- Designed and implemented five end-to-end RAG pipelines from naive dense retrieval to graph-based, agentic, and LangChain LCEL architectures, with a provider-agnostic interface supporting Anthropic API, OpenAI API, and Ollama.
- Implemented hybrid retrieval (BM25+dense search, RRF fusion, cross-encoder reranking, HyDE), knowledge graph construction, and agentic reasoning (Self-RAG, CRAG).

Vision Fairness Analysis — CNN vs Vision Transformer

Berkeley, California

Machine Learning Engineer, **UC Berkeley**

Aug 2025 – Dec 2025

- Designed a controlled fairness study comparing U-Net encoders (ResNet50 vs. FastViT, both ImageNet-1k pretrained) on $4\times$ super-resolution and racial classification across a 7-group FairFace dataset (108k images) and 4-group RFW dataset, with a **1:20 minority-to-majority imbalance** simulation per race.
- Fine-tuned backbones via staged layer unfreezing (ResNet50: layer4→layer3; FastViT: layers 5–7→3–4) using VGG19-based perceptual loss for upscaling and cross-entropy for classification; ResNet50 achieved **+5.79 dB higher PSNR** than FastViT,

while both models showed ≤ 0.05 dB per-race PSNR variance across all imbalance settings, confirming super-resolution is race-agnostic.

- Quantified classification fairness via a per-group representation ratio ($\text{Acc}(r)/\text{AvgAcc}$); identified Indian and Caucasian groups as most sensitive to underrepresentation (-18% and -16% accuracy drops), while African faces remained robust (-3% drop) under heavy downsampling.
- Found FastViT more resilient to demographic imbalance (10.37% avg. accuracy drop vs. ResNet50's 14.13%); near-identical per-race trends across architectures indicate fairness degradation is driven primarily by training data distribution rather than model architecture.

Smart Parking

Robotics Engineer, NTU

Taipei, Taiwan

Aug 2023 – Jan 2023

- Collaborated with a teammate to develop a smart driving system, featuring real-time notifications and auto-parking.
- Integrated PiRacer Pro and STM32 board to boost system capabilities with analytics and imitation learning.

PUBLICATIONS & CONFERENCE EXPERIENCE

- **Co-First Author, COLM 2024** — Li-Chun Lu*, **Shou-Jen Chen***, Tsung-Min Pai, Chan-Hung Yu, Hung-yi Lee, Shao-Hua Sun, "LLM Discussion: Enhancing the Creativity of Large Language Models via Discussion Framework and Role-Play" 
- **Co Author, ICLR 2025** — Chien-yu Huang et al., **Shou-Jen Chen...**, Hung-yi Lee, "Dynamic-SUPERB Phase-2: An Open Benchmark Evolving through Collaborative Expansion for Comprehensive Evaluation of Spoken Language Models with 180+ Tasks" 
- **Reviewer, 2024 IEEE Spoken Language Technology Workshop (SLT)**, Reviewed and evaluated research paper.

RESEARCH EXPERIENCE

Robot Learning Laboratory (RLL), NTU

Taipei, Taiwan

Research Assistant / Undergraduate Researcher | Advisor: **Prof. Shao-Hua Sun**

Oct 2023 – Jun 2025

- Introduced LLM Discussion framework that enhances the creativity of large language models via multi-agent discussion and role-playing, surpassing single-agent and existing multi-agent LLM frameworks in creativity tests by 20%.
- Designed an LLM evaluation focused on creativity metrics, achieving a 0.7 correlation with human on 1,400+ responses.
- Develop a decision-making agent that optimizes selection between in-context learning with Retrieval-Augmented Generation, supervised fine-tuning, and reinforcement learning from human feedback in a streamline learning process.
- Investigated test-time compute scaling by measuring the correlation between algorithmic time complexity (Big-O) and the number of tokens required for LLMs to solve algorithmic problems.

Speech Processing and Machine Learning Laboratory (SPMLLab), NTU

Taipei, Taiwan

Undergraduate Researcher | Advisor: **Prof. Hung-yi Lee**

Jul 2023 – Jun 2025

- Devised and preprocessed a "SoundEffectDetection" dataset/task for Dynamic-SUPERB, aimed at building universal speech models capable of leveraging instruction tuning to perform multiple tasks in a zero-shot fashion. 
- Utilize Direct Preference Optimization (DPO) as a method for Reinforcement Learning from Human Feedback (RLHF) to train a text-instruction-guided voice conversion model, aiming to enhance model performance.

Wireless and Mobile Networking Laboratory (WMNLab), NTU

Taipei, Taiwan

Undergraduate Researcher | Advisor: **Prof. Hung-Yu Wei**

Jan. 2023 – Jan. 2024

- Conducted mobile phone experiments on the MRT using user datagram protocol to evaluate network handover.
- Analyzed data and presented findings on experiments across a 25-km route to professors and graduate students.

OTHER EXPERIENCE

Generative Artificial Intelligence Course, NTU (Technical Report / Course Webpage

Taipei, Taiwan

Teaching Assistant (TA)

Feb. 2024 – Jun. 2024

- Collaborated with 30 TAs to manage and respond to student emails for a class of over 1,000 students and 1,000 auditors.
- Designed and graded a coursework related to calling API alongside two other TAs and provided 8 office hours.

English Written and Oral Communication for Professional Pursuit Course, NTU

Taipei, Taiwan

Teaching Assistant (TA)

Sept. 2024 – Jan. 2025

- Dedicate over 30 hours per month throughout the semester to support course instruction and weekly reminders.
- Manage and promptly respond to emails and inquiries for a class of over 70 students, ensuring timely communication.